Karthik Shivashankaran

Brooklyn, NY | +1 848-702-5284 | ks7926@nyu.edu

imaginelenses.com | linkedin.com/in/imaginelenses | github.com/imaginelenses

US Work Authorized — OPT (STEM Extension Eligible)

SUMMARY

M.S. Computer Engineering (2026) specializing in Robotics and Autonomous Navigation. Built and deployed full autonomy stacks on physical hardware, from perception through planning; implemented state estimation and control algorithms from scratch. Open-source creator with 130+ GitHub stars.

EDUCATION

New York University — M.S. Computer Engineering · Scholarship Recipient New York, NY | 2026

Courses: Robot Localization & Navigation, Reinforcement Learning for Robotics, Computer Vision, , Real-Time Embedded Systems, Computer Architecture, Operating Systems

Acharya Institute of Technology — B.E. Mechatronics Bangalore, India | 2022

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL, CMake, MATLAB

ML / Computer Vision: YOLOv8, Object Detection, Multi-Object Tracking, ByteTrack, 3D Gaussian Splatting, Depth & Pose Estimation, Structure-from-Motion (SfM), Camera Calibration, PyTorch, OpenCV, NumPy

Robotics & Autonomy: SLAM, Motion Planning, Path Planning, Sensor Fusion, State Estimation (Kalman Filter, EKF, Particle Filter), Open3D, Point Cloud Processing, Reinforcement Learning (Stable-Baselines3), Gymnasium, ROS2 (familiar)

Control Systems: PID, MPC, LQR, MPPI, Closed-Loop Control, Real-Time Systems

Embedded & Hardware: ESP32, Arduino Giga R1, Raspberry Pi, IMU, Circuit Design, CAD, 3D Printing

Systems & Tools: Git, Docker, CI/CD, Linux, AWS (EC2, S3, Lambda), TCP/IP, WebSockets, CAN, UART, SPI, I2C

APPLIED RESEARCH & TECHNICAL PROJECTS

pixelsToSteps · *Python · NumPy · SciPy · OpenCV · Gymnasium* May 2026

- Achieved 100% survival across 100 trials under true plant mismatch (pole mass ×1.1, half-length ×0.9) stabilizing a cartpole from raw pixels alone; used an LQR teacher and hybrid Luenberger observer (Ridge regression, 40 demos), up from 96/100 at half the data
- Observer maps Gaussian-blur + Otsu-thresholded frames to pole angle ($R^2 = 0.987$, RMSE = 0.123°), fusing pixel estimates with linearized dynamics at 60 Hz as a single matrix multiply with no angle sensor at inference

Autonomous Robot Navigation Stack · *Python · C++ · UDP · RPLidar · ArUco · SciPy* Apr 2026

- Closed the full autonomy loop on a physical Ackermann robot (Arduino Giga R1 + ESP32 for motor control and sensing, compute offloaded to host over UDP) with a from-scratch MPPI controller; sampled thousands of stochastic trajectories per step, weighted by an EDT-based composite cost (goal proximity + obstacle clearance), achieving real-time autonomous navigation
- Fused UDP-streamed wheel encoder odometry (10 Hz) with ArUco visual pose measurements via a from-scratch EKF, reducing positional RMSE to 4.14 cm and recovering from pose errors in <0.1 s
- Built a 3,000-particle Monte Carlo Localizer for global localization against a pre-built LiDAR occupancy map; achieved 0.27 cm position spread and 0.90° heading σ, recovering from a 1 m / 90° initialization error in 0.63 s

3D Line-of-Sight Audit System for Intersections · *Python · Gaussian Splatting · OpenCV* Dec 2025

- Owned the 3D reconstruction pipeline within a 5-person team-built intersection audit system, replacing manual checks with an 8-camera SfM and Gaussian Splatting pipeline (30 FPS); generated a 1.34M-point cloud to evaluate pedestrian visibility
- Contributed to deployment of YOLOv8 + ByteTrack for real-time multi-object tracking (20–40 tracks/ camera); multi-camera fusion achieved >90% precision, >85% recall, and detected vehicles ∼12.2s earlier than a single-camera 2D baseline

Smart Switch · *JavaScript · YAML · ESP32 · Circuit Design · CAD · 3D Printing* Aug 2023

- Owned end-to-end from circuit design and 3D-printing through firmware and auto-generated YAML config tooling; eliminated cloud round-trip latency (sub-ms LAN vs. 80–200 ms cloud), 5 units deployed and operational

Blendit (★ 130 GitHub Stars) · *Python · Git · Blender* Sep 2022

- Git-based version control extension for Blender; reduced project file size by 83% via function-call delta logging instead of full binary snapshots